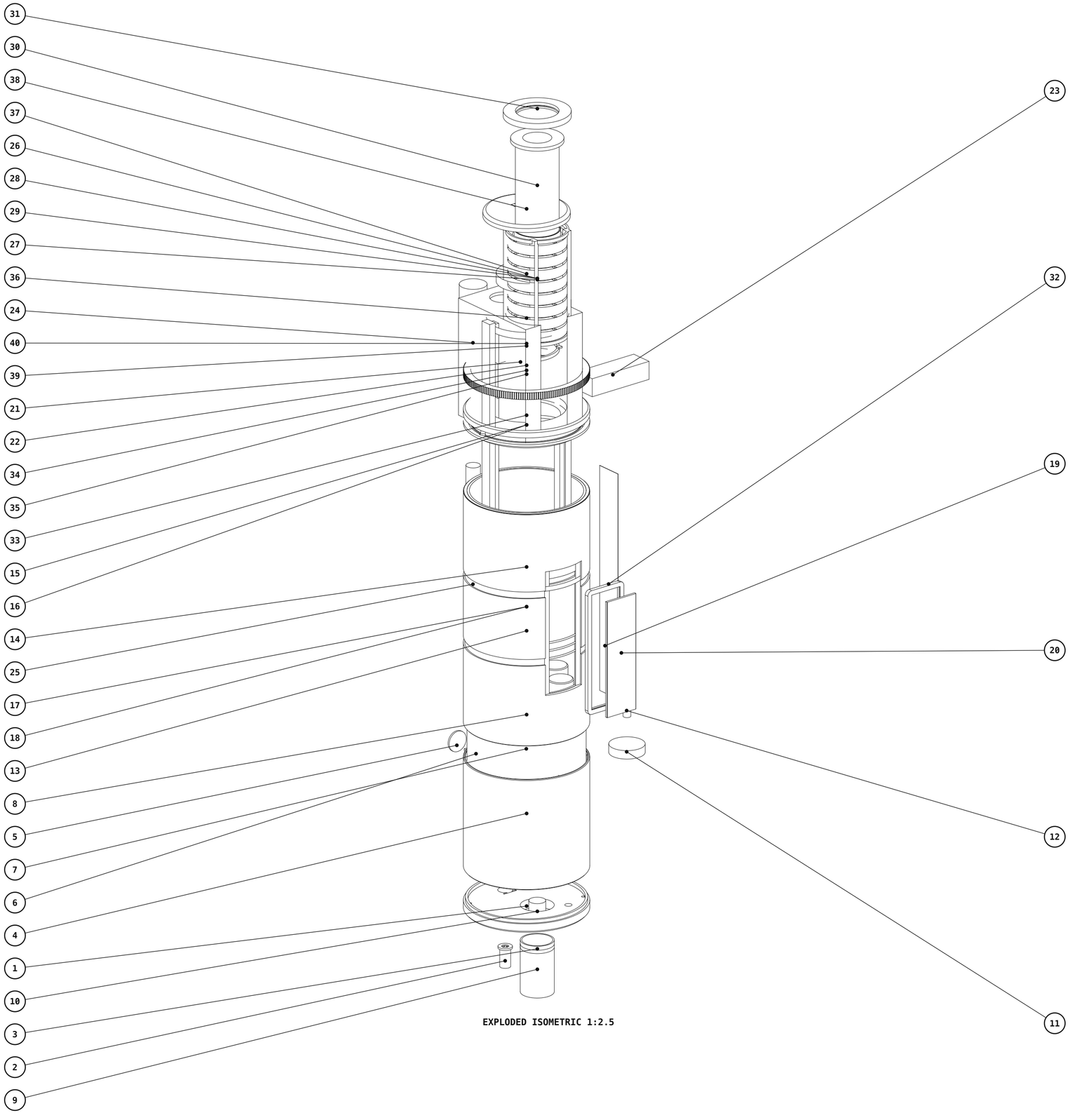
WIP - ENGINEERING REVIEW ONLY - NOT RELEASED FOR FABRICATION



EXPLODED ISOMETRIC 1:2.5

ITEM	PART NO.	DESCRIPTION	QTY	MATERIAL	KIND	
1	SBL-CO-001	Base cap	1	Ti-6Al-4V ELI, ASTM F136	custom	
2	SBL-CO-B01	LEMO K-series 1B receptacle (IP68)	1	Bought component (datasheet envelope	bought	
3	SBL-CO-027	Lube capsule bayonet door	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
4	SBL-CO-002	Base ring	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
5	SBL-CO-029	Avionics eye window Ø18	1	C-plane sapphire, AR both sides	custom	
6	SBL-CO-B02	Status LED flex	1	Polyimide flex, 2-layer, ENIG pads	bought	
7	SBL-CO-003	Cell carrier	1	PEEK (Victrex 450G, black)	custom	
8	SBL-CO-B03	21700 cell ×6 (6S1P, Molicel P45B)	6	Li-ion 21700 cell (Molicel P45B)	bought	
9	SBL-CO-026	Lube capsule	1	PEEK (Victrex 450G, black)	moulded	
10	SBL-CO-B04	Lube micropump	1	Bought component (datasheet envelope	bought	
11	SBL-CO-B05	Suction disc pump	1	Bought component (datasheet envelope	bought	
12	SBL-CO-B06	Vacuum relief valve (Lee CKFA)	1	Bought component (datasheet envelope	bought	
13	SBL-CO-B07	Avionics stack (safety / compute / power boards)	3	FR-4 PCB, 1.6 mm, black soldermask	bought	
14	SBL-CO-004	Bulkhead	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
15	SBL-CO-005	Spine bar ×2	2	Ti-6Al-4V, ASTM B348 Gr 5	custom	
16	SBL-CO-B08	HIWIN MGN7 rail ×2	2	Stainless 304L, ASTM A276	bought	
17	SBL-CO-006	Structural tube	1	PEEK 30% CF (Victrex 450CA30)	custom	
18	SBL-CO-007	Exterior skin	1	Pt-cure silicone gel, Shore 00-10/00	moulded	
19	SBL-CO-008	Window bezel	1	Ti-6Al-4V ELI, ASTM F136	custom	
20	SBL-CO-009	HASEL window	1	C-plane sapphire, AR both sides	custom	
21	SBL-CO-016	Carriage frame	1	PEEK 30% CF (Victrex 450CA30)	custom	
22	SBL-CO-B09	HIWIN MGN7H block ×2	2	Stainless 304L, ASTM A276	bought	
23	SBL-CO-023	HV pod (potted)	1	PEEK (Victrex 450G, black)	custom	
24	SBL-CO-B10	LinMot PS02-23Sx80F stator (moving)	1	Bought component (datasheet envelope	bought	
25	SBL-CO-B11	LinMot slider rod Ø12 (fixed)	1	Stainless 304L, ASTM A276	bought	
26	SBL-CO-018	Cassette cage	1	PEEK 30% CF (Victrex 450CA30)	custom	
27	SBL-CO-019	HASEL sector ×32 (8 rings × 4)	32	Polyimide flex, 2-layer, ENIG pads	bought	
28	SBL-CO-022	Ground screen	1	Carbon-loaded conductive silicone	moulded	
29	SBL-CO-021	Sensing membrane	1	Medical LSR, Shore A20-A40	moulded	
30	SBL-CO-024	Liner (removable)	1	Pt-cure silicone gel, Shore 00-10/00	moulded	
31	SBL-CO-016a	Carriage top collar	1	Ti-6Al-4V ELI, ASTM F136	custom	
32	SBL-CO-025	Rolling flex	1	Polyimide flex, 2-layer, ENIG pads	bought	
33	SBL-CO-010	Cap ring	1	Ti-6Al-4V ELI, ASTM F136	custom	
34	SBL-CO-011	STOP collar	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
35	SBL-CO-012	Light ring	1	Optical silicone light guide	custom	
36	SBL-CO-013	Lube/UV bezel	1	3Y-TZP zirconia, ivory	custom	
37	SBL-CO-014	Iris lip	1	Pt-cure silicone gel, Shore 00-10/00	moulded	
38	SBL-CO-015	Dust lid	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
39	SBL-CO-028	UV window ring	1	C-plane sapphire, AR both sides	custom	
40	SBL-CO-B12	UV-C LED ×4	4	Bought component (datasheet envelope	bought	

1. Explode offsets are illustrative; items balloon to the BOM.
2. Bought components are datasheet envelopes (kind = bought).

SABLE ROBOTICS — **SBL-CO**

EXPLODED ASSEMBLY DRAWING · Sable Core — **REV A**

MATERIAL	FINISH	SCALE
see BOM	see BOM	
GENERAL TOLERANCE	SURFACE	MASS
ISO 2768-mK · UNITS mm · ANGLES °	see Ra callouts; unmarked edges 0.6 × 45°	—
THIRD ANGLE PROJECTION	SHEET 2 / 6	A2 · 2026-11-08 · parts/cad/src